A First Look at Machine Learning

Engineering Computation with Python — Chapter 15

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The Computer Learns the Rule

An enrichment chapter

This chapter is not a separate teaching week. It shows the main ideas with two very small models.

Traditional programming

A person writes the rule or equation that connects inputs to outputs.

Machine learning

A model is trained on example inputs and outputs, and learns the rule itself.

Three Types of Machine Learning

Supervised	Unsupervised	Reinforcement
Learns from labelled training data. *E.g.* regression, classification.	Learns from unlabelled data, finding patterns on its own.	Learns through feedback, reinforcing correct responses.

This chapter's examples are all supervised — regression and decision-tree classification.

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The General Machine Learning Workflow



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```python
import pandas as pd
from sklearn.preprocessing import StandardScaler

df = pd.DataFrame({'height': [157, 163, 176, 152, None]})
df['height'] = df['height'].fillna(df['height'].mean())   # cleaning
df['height_scaled'] = StandardScaler().fit_transform(df[['height']])
print(df)
```

Fit the scaler on the training data only

When you also split the data into training and testing sets, fit the scaler on the training set and use it to transform both. Otherwise the test set leaks into training.

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Regression vs. Classification

Predictive models divide into two kinds, by what they produce:

Regression

Predicts a quantity — a temperature, a voltage, a time constant.
"How hot will this motor get?"

Classification

Predicts a category — healthy or faulty, pass or fail.
"Is this motor overheating?"

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A Simple Decision Tree

```
from sklearn.tree import DecisionTreeClassifier

X = [[1], [2], [3], [4]]
y = ['low', 'low', 'high', 'high']

model = DecisionTreeClassifier()
model.fit(X, y)
print(model.predict([[3]])[0])   # high
```

A model has no sense of its own limits

Asked to classify 99 — far outside its training data — it still returns `high` regardless, since its only rule is “above 2.5”. Judging when a question falls outside the model’s experience remains the engineer’s job.

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A Straight-Line Regression

```python
import numpy as np
from sklearn.linear_model import LinearRegression

X = np.array([26, 30, 35, 40]).reshape(-1, 1)   # temperature (C)
y = np.array([1000, 2500, 3500, 4500])          # ice-cream sales

model = LinearRegression().fit(X, y)
print(model.predict([[45]])[0])                 # 5847.62979684...
```

This is an extrapolation

45 °C is outside the training range (26 to 40 °C). The model just extends its line; nothing in the data shows that sales keep rising the same way. Trust it much less than a prediction inside the range.

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Hold Some Data Back

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X = [[22], [24], [26], [28], [30], [32], [34], [36], [38], [40]]
y = [450, 700, 1000, 1700, 2500, 2800, 3300, 3700, 4100, 4500]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0)
model = LinearRegression().fit(X_train, y_train)
print(f"{model.score(X_train, y_train):.3f}")   # 0.992 (training)
print(f"{model.score(X_test, y_test):.3f}")     # 0.975 (unseen data)
```

The test score is the honest one

A large gap between the two scores is the classic sign of overfitting.

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Where This Leaves You

- In machine learning a model learns the rule from examples; this book uses only supervised learning.
- `DecisionTreeClassifier` predicts a category; `LinearRegression` predicts a number.
- Always judge a model on a test set it has not seen, and fit any scaler on the training set only.
- No model can be trusted outside the range of its training data: machine learning is one more engineering tool, not a substitute for understanding the problem.

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